Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the test accuracy by dataset.

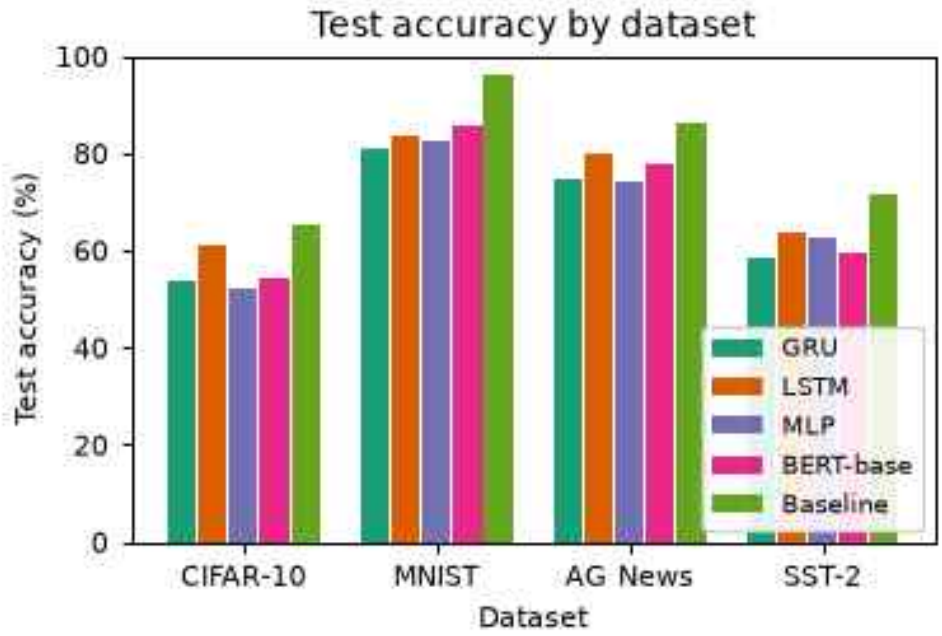


Figure 1: Test accuracy by dataset for the models in this experiment.

Baseline achieves 74.2% test accuracy on AG News. Baseline achieves 71.5% test accuracy on SST-2. BERT-base reaches 78% test accuracy on AG News (Figure 1). On MNIST, LSTM is 12.5 points better than GRU.

Figure 2 shows the f1 score by dataset.

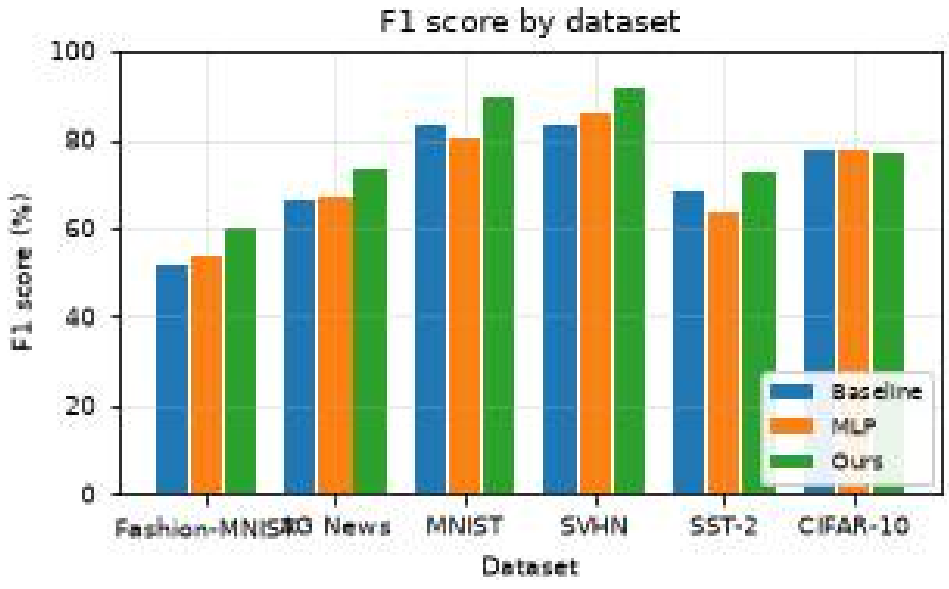


Figure 2: F1 score by dataset for the models in this experiment.

Ours reaches over 70% F1 score on SST-2 (Figure 2). On Fashion-MNIST, Ours is 8.5 points better than

Baseline.

Figure 3 shows the test accuracy during training.

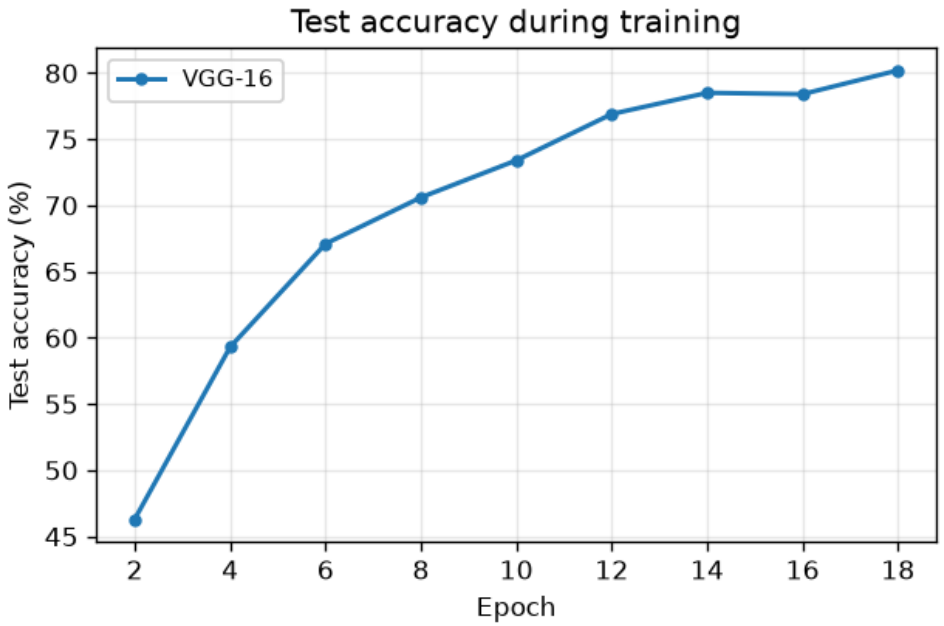


Figure 3: Test accuracy during training for the models in this experiment.

As shown in Figure 3, VGG-16 attains a test accuracy of 66.9% at epoch 8. VGG-16 attains a test accuracy of 78.8% at epoch 12. VGG-16 scores 1.5 points higher on 16 than on 12 (Figure 3). VGG-16 attains a test accuracy of 59% at epoch 4.

Figure 4 shows the accuracy during training.

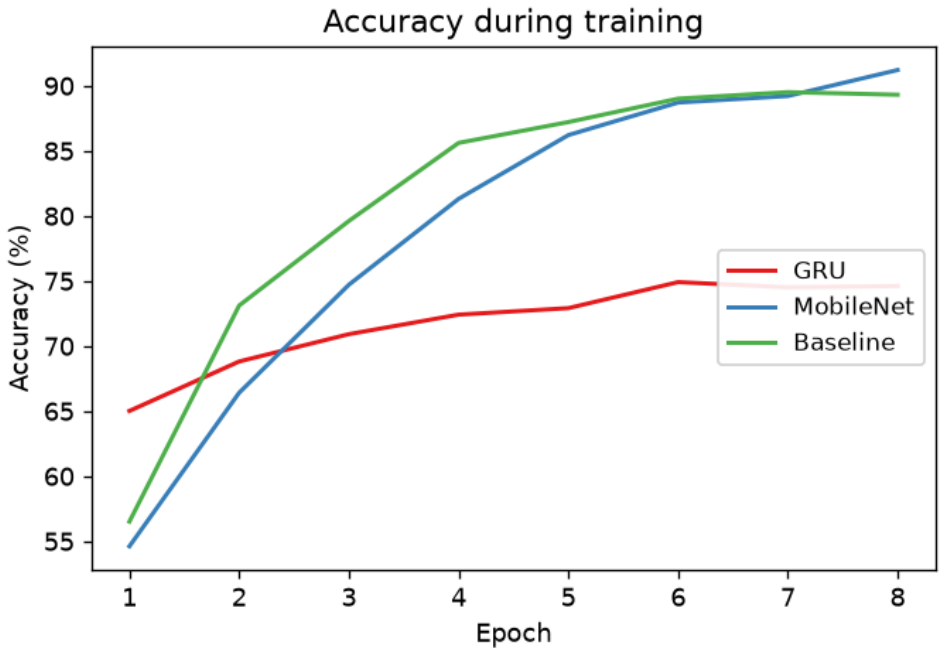


Figure 4: Accuracy during training for the models in this experiment.

As shown in Figure 4, MobileNet outperforms GRU by 23.9 percentage points at epoch 3. MobileNet outperforms GRU by 9 percentage points at epoch 4.

3 Discussion

On STL-10, SVM obtains an accuracy of 79.1%. Overall, the results are in line with what we expected from

the literature.